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ECE 5880 / CHEME 5880, Energy Engineering Seminar I
Seminar summary, September 10, 2026

Todd DuMond spoke about running a farm as a circular economy, where crop production, soybean processing, animal feed, fuel and electricity each feed the next step instead of being separate businesses. His stated goal is a system that produces large amounts of food, feed and energy at close to no outside energy input, and possibly at negative carbon. What made the talk useful was that he kept tying that vision back to specific equipment and specific numbers rather than leaving it abstract.

He walked through how the business got here. In 2018 they organized about twenty dairy farms to supply non-GMO milk for a Fage yogurt plant, which he described as finding a differentiated specialty product inside a commodity industry. The soybean opportunity came from a similar observation. New York grows very good soybeans, but they were being containerized out of the Port of New York while local dairies were buying protein in, and the farmer was absorbing the basis. So in 2020 they built an extrude and expel plant. Beans enter the extruder at roughly 100 degrees and come out near 350 degrees about thirty seconds later, which ruptures the cells and vaporizes the internal water, and a press then removes two thirds to three quarters of the oil. The feed product runs around 48% protein at about 6% fat with the fiber intact, and the soybean oil goes to biodiesel and other uses.

The part I found most interesting was his argument that biology can substitute for diesel. Two pounds of rye seed planted in the fall will deliver about twenty pounds of sulfur to next year's crop, so he does not buy sulfur. A radish sends a root down four feet, which does roughly the work of ripping the soil sixteen inches with a large tractor, except that the radish costs no fuel. He also pointed out that nitrogen uptake is concentrated, with something like 80% taken up in about a month, so applying everything early mostly creates an opportunity for it to wash away. In each case the efficiency gain is not a better machine but a decision to stop using a machine.

On the energy side, they are installing a one to two megawatt solid oxide fuel cell within twelve months, fueled by renewable natural gas from a nearby dairy digester. He gave it as 54% efficient at making electricity, which he compared favorably to a new gas turbine, with another 36% of the input energy leaving as roughly 800 degree heat. Because they can use all of that heat in the crush plant, which is otherwise burning a great deal of electricity purely to make heat, the combined efficiency gets close to 90% and directly displaces natural gas or propane.

He was also candid about what does not work. He ended up owning a biodiesel company mostly because a customer could not pay him, and his repeated point was that biodiesel demand is created by the RVO, 45Z and low carbon fuel standards rather than by economics, and that a policy change can destroy the business. His stated aim is to attach biodiesel to the crush plant and power it with the fuel cell so that it pencils without the subsidy. He said something similar about cover crops, that they cover 60 to 85% of their acres and earn about \$70 an acre through carbon programs, but that flying seed on after grain corn in November is done to get paid rather than because it works, and he would rather use practices that actually do something than check a box.

What I took from the seminar is that the constraint here is rarely the science. The agronomy, the fuel cell and the transesterification chemistry are all well understood. What decides whether any of it gets built is whether the policy holds and whether a consumer will pay more for a carbon negative egg. That is the third seminar in a row where the technology was settled and the uncertainty was somewhere else.